

# $T\bar{T}$ deformation in large $N$ limit

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Based on work in progress with T Ishii, H Kawai, K Sakai, K Yoshida

# Introduction

- ◇  $T\bar{T}$  deformation is an **irrelevant** deformation of 2D (classical) Lagrangian with the  $T\bar{T}$  flow equation:

$$\frac{\partial}{\partial \alpha} \mathcal{L}^{(\alpha)} = \det \left( T_{\mu\nu}^{(\alpha)} \right)$$

where  $T_{\mu\nu}^{(\alpha)}$  is the energy-momentum tensor defined from  $\mathcal{L}^{(\alpha)}$ .

[Smirnov-Zamolodchikov, Cavaglià-Negro-Szécsényi-Tateo]

- ◇ This is studied in **various contexts**.  
(Integrable QFT and integrability, coupling to topological gravity, Higher-dimensional generalization, Holography, Little string theory ...)

# Introduction

- ◇ Classical aspects of the  $T\bar{T}$  flow have been studied well.  
But **quantum** aspects of the  $T\bar{T}$  flow are **not**.
  - ◇ Some researchers have pointed out that  
 $T\bar{T}$  deformation is equivalent to classical metric perturbation.  
[Cardy 2018, Dubovsky-Gorbenko-Mirbabayi 2017, Conti-Negro-Tateo 2018]
- ◇ If the  $T\bar{T}$  flow can be understood at **quantum** level,  
a clue to **quantum gravity** may be captured.
- ◇  $T\bar{T}$  deformation is hard to study by perturbative methods.  
⇒ We use **large N** method.

# What I will talk about

- ◆ We study infinitesimal  $T\bar{T}$  deformation of **massive  $O(N)$  vector model** in the large  $N$  limit.

$$\frac{\mathcal{L}}{N} = \frac{1}{2} (\partial_\mu \vec{\phi})^2 - \frac{m_0^2}{2} \vec{\phi}^2 + \alpha_0 \det(T_{\mu\nu})$$

- ◆ We find that

1. Divergence can be **renormalized** (up to  $O(1/N)$ ).
2. **Positivity** of state space is **violated**.
3. **Diffeomorphism** invariance is **needed** to recover the positivity.
4. Above results **holds** for **general** deformations with **arbitrary** potential.

# $T\bar{T}$ in large N limit

- ◆ Consider the following Lagrangian:

$$\frac{\mathcal{L}}{N} = \frac{1}{2} (\partial_\mu \vec{\phi})^2 - \frac{m_0^2}{2} \vec{\phi}^2 + \alpha_0 \det(T_{\mu\nu})$$
$$T_{\mu\nu} := \partial_\mu \vec{\phi} \cdot \partial_\nu \vec{\phi} - \frac{\eta_{\mu\nu}}{2} \left[ (\partial_\rho \vec{\phi})^2 - m_0^2 \vec{\phi}^2 \right]$$

- ◆ This Lagrangian is a solution to the  $T\bar{T}$  flow equation with “initial condition” of

$$\frac{\mathcal{L}}{N} = \frac{1}{2} (\partial_\mu \vec{\phi})^2 - \frac{m_0^2}{2} \vec{\phi}^2$$

up to  $O(\alpha_0)$ . [Bonelli-Doroud-Zhu]

- ◆ We calculate **the effective action** in  $N \rightarrow \infty$  limit.

# $T\bar{T}$ in large N limit

◊ Rewriting  $\mathcal{L}$  with auxiliary fields  $C, \tilde{C}^{\mu\nu}$ :

$C$  ... scalar field  
 $\tilde{C}^{\mu\nu}$  ... traceless symmetric tensor field

$$\frac{\mathcal{L}}{N} = \frac{1}{2}(\partial_\mu \vec{\phi})^2 - \frac{m_0^2}{2}\vec{\phi}^2 - \frac{m_0^2}{2}C\vec{\phi}^2 - \frac{1}{2}\tilde{C}^{\mu\nu}\partial_\mu \vec{\phi} \cdot \partial_\nu \vec{\phi} - \frac{1}{4\alpha_0}C^2 + \frac{1}{8\alpha_0}(\tilde{C}^{\mu\nu})^2$$

(With E.O.M of  $C, \tilde{C}^{\mu\nu}$ , this reproduces the original  $\mathcal{L}$  )

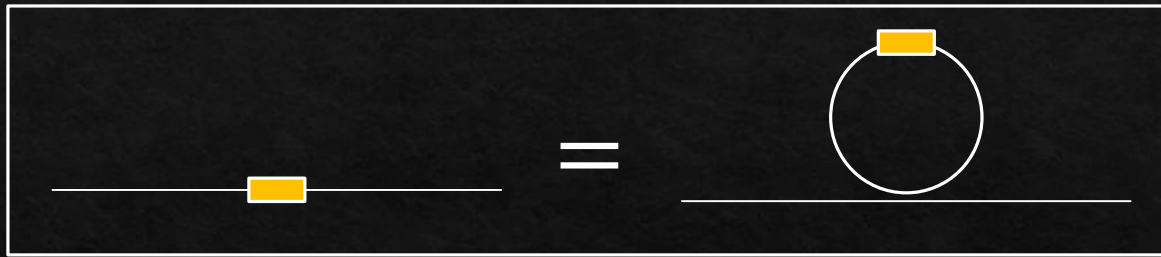
◊ The effective action:

$$\begin{aligned} \frac{S_{\text{eff}}}{N} = & \frac{i}{2} \text{tr} \log(-\partial^2 - (1 + C)m_0^2 + 2\partial_\mu \tilde{C}^{\mu\nu} \partial_\nu) \\ & + \int_x \frac{-1}{4\alpha_0} C^2 + \frac{1}{8\alpha_0} (\tilde{C}^{\mu\nu})^2 \end{aligned}$$

# $N \rightarrow \infty$ limit

- ◇ In  $N \rightarrow \infty$  limit, the stationary point of  $S_{\text{eff}}$  dominates for  $C$  &  $\tilde{C}^{\mu\nu}$  integral.
- ◇ Stationary condition (E.O.M or Gap equation) for  $C$ :

$$\frac{1}{\alpha_0 m_0^2} \langle C \rangle = \frac{1}{4\pi^2 i} \int_k \frac{d^2 k}{k^2 - (1 + \langle C \rangle) m_0^2}$$



- The logarithmic divergence of R.H.S can be renormalized into  $\alpha_0 m_0^2$ .
- ◇ From translation and Lorentz invariance,  $\langle \tilde{C}^{\mu\nu} \rangle = 0$ . This satisfies the stationary condition trivially.

# 1/N correction

- ◇ To consider  $\frac{1}{N}$  (**sub-leading**) contributions,  
 we set  $C = \langle C \rangle + \frac{m^2}{m_0^2} \tilde{C}$  and expand  $S_{\text{eff}}$  up to **2<sup>nd</sup> order of  $\tilde{C}, \tilde{C}^{\mu\nu}$** :

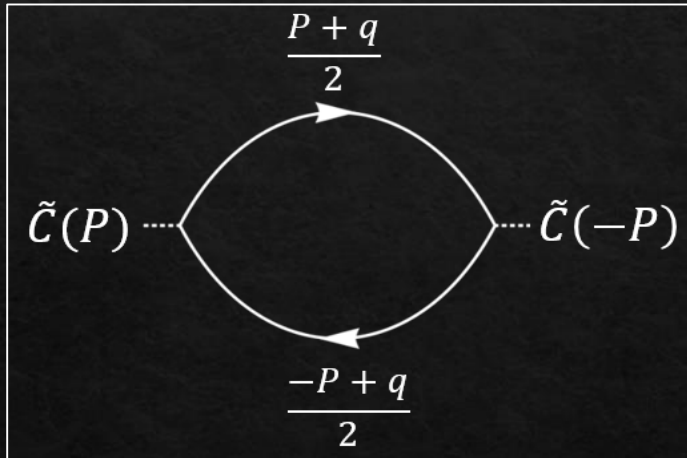
$$\frac{S_{\text{eff}}}{N} = \frac{1}{2} \int_{x,y} (\tilde{C}_{(x)} \tilde{C}_{(x)}^{\alpha\beta}) \begin{pmatrix} M & M_{\mu\nu} \\ M_{\alpha\beta} & M_{\mu\nu\alpha\beta} \end{pmatrix} \begin{pmatrix} \tilde{C}_{(y)} \\ \tilde{C}^{\mu\nu}_{(y)} \end{pmatrix}$$

$$\left[ \begin{aligned} M &= \frac{m^4}{2i} (G(x-y))^2 - \frac{1}{2\alpha_0} \delta(x-y) \\ M_{\mu\nu} &= \frac{m^2}{2i} \partial_\mu G(x-y) \partial_\nu G(x-y) \\ M_{\mu\nu\alpha\beta} &= \frac{1}{2i} \partial_\mu \partial_\alpha G(x-y) \partial_\nu \partial_\beta G(x-y) + \frac{\eta_{\mu\alpha} \eta_{\nu\beta}}{4\alpha_0} \delta(x-y) \\ G(x-y) &:= (-\partial^2 + (1 + \langle C \rangle) m_0^2)^{-1} \end{aligned} \right]$$

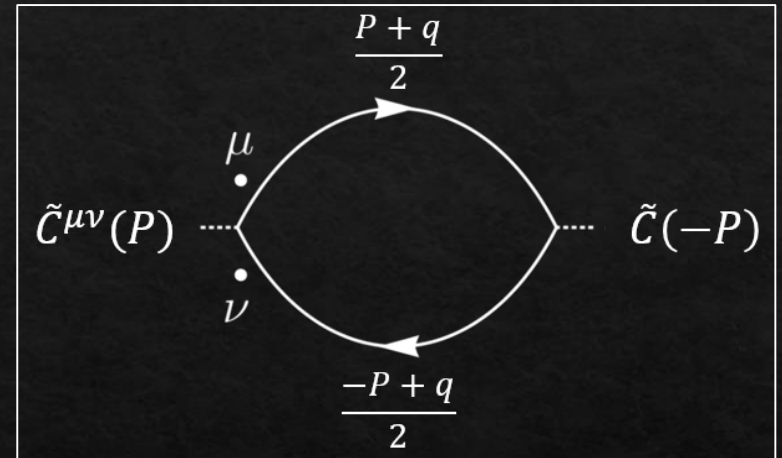
# 1/N correction

◇  $M, M_{\mu\nu}, M_{\mu\nu\alpha\beta}$  correspond the following diagrams:

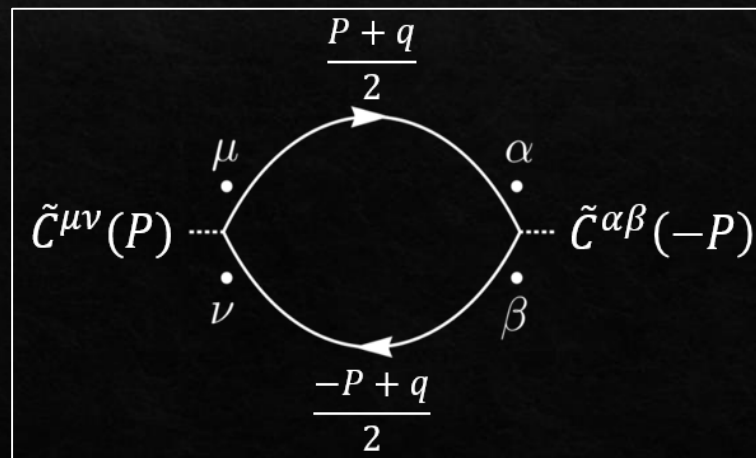
$M$



$M_{\mu\nu}$



$M_{\mu\nu\alpha\beta}$



# 1/N correction

$$m^2 := (1 + \langle C \rangle) m_0^2$$

◆ After some calculation, we get

$$\begin{aligned} \frac{S_{\text{eff}}}{N} = & \frac{1}{2} \int_P \left( a P^2 - \frac{1}{4\alpha_0} + (\text{quadratic div.}) \right) (\tilde{C}^{\mu\nu})^2 \\ & + \left[ \left( \frac{m^4}{4\pi P \sqrt{4m^2 - P^2}} \right) \tan^{-1} \left( \frac{P}{\sqrt{4m^2 - P^2}} \right) - \frac{P^2}{96\pi} - \frac{m^2}{16\pi} \right] \left( \tilde{C} + \frac{P_\mu P_\nu}{P^2} \tilde{C}^{\mu\nu} \right)^2 \end{aligned}$$

$$\left( \begin{array}{l} a \cdots \text{regularization-dependent constant} \\ \text{e.g. } 0 \text{ (Pauli-Villars, Dimensional)} \\ \frac{-1}{12\pi} \text{ (Cutoff)} \end{array} \right)$$

# Renormalizability

$$\frac{S_{\text{eff}}}{N} = \frac{1}{2} \int_P \left( aP^2 - \frac{1}{4\alpha_0} + (\text{quadratic div.}) \right) (\tilde{C}^{\mu\nu})^2$$

$$+ \left[ \left( \frac{m^4}{4\pi P \sqrt{4m^2 - P^2}} \right) \tan^{-1} \left( \frac{P}{\sqrt{4m^2 - P^2}} \right) - \frac{P^2}{96\pi} - \frac{m^2}{16\pi} \right] \left( \tilde{C} + \frac{P_\mu P_\nu}{P^2} \tilde{C}^{\mu\nu} \right)^2$$

◇ The divergence of mass term (quadratic divergence) can be renormalized into  $\alpha_0$

⇒ **Divergences** of the gap equation and the 1-loop effective action can be **renormalized** despite of irrelevant deformation!

◇ The coefficient of  $P^2(\tilde{C}^{\mu\nu})^2$  is **finite** but **regularization dependent**.

Finite renormalization corresponds to adding **non-local** counterterm:

$$\frac{\delta_Z}{2} \tilde{C}^{\mu\nu} \partial^2 \tilde{C}_{\mu\nu} \leftrightarrow 2\alpha_0^2 \tilde{T}^{\mu\nu} \frac{\delta_Z \partial^2}{1 + 4\alpha_0 \delta_Z \partial^2} \tilde{T}_{\mu\nu} \in \mathcal{L}_{\text{c.t.}}$$

# Positivity

$$\frac{S_{\text{eff}}}{N} = \frac{1}{2} \int_P \left( aP^2 - \frac{1}{4\alpha_0} + (\text{quadratic div.}) \right) (\tilde{C}^{\mu\nu})^2$$

$$+ \left[ \left( \frac{m^4}{4\pi P \sqrt{4m^2 - P^2}} \right) \tan^{-1} \left( \frac{P}{\sqrt{4m^2 - P^2}} \right) - \frac{P^2}{96\pi} - \frac{m^2}{16\pi} \right] \left( \tilde{C} + \frac{P_\mu P_\nu}{P^2} \tilde{C}^{\mu\nu} \right)^2$$

◇  $\tilde{C}$  &  $\tilde{C}^{\mu\nu}$  have kinetic terms with **wrong sign**.

1. In  $P^2 \rightarrow \infty$  region, the second term behaves like  $-\frac{P^2}{96\pi}$
2. If  $a \neq 0$ , the first term becomes

$$aP^2 (\tilde{C}^{\mu\nu})^2 = aP^2 (\tilde{C}^{00})^2 - aP^2 (\tilde{C}^{01})^2$$

$\Rightarrow$  **Positivity** of the state space is **violated**!

◇ What is necessary to recover the positivity?

# Liouville action

◇ The rewritten Lagrangian:

$$\frac{\mathcal{L}}{N} = \frac{1}{2} \left( (\eta^{\mu\nu} - \tilde{C}^{\mu\nu}) \partial_\mu \vec{\phi} \cdot \partial_\nu \vec{\phi} - m^2 (1 + \tilde{C}) \vec{\phi}^2 \right) + (\text{mass terms})$$

◇ This is similar to (massive) Polyakov action

$$\mathcal{L} = \sqrt{-g} (g^{\mu\nu} \partial_\mu \vec{\phi} \cdot \partial_\nu \vec{\phi} - m^2 \vec{\phi}^2)$$

regarding

$$g_{\mu\nu} = \hat{g}_{\mu\nu} e^\sigma \leftrightarrow \eta_{\mu\nu} + \tilde{C} \eta_{\mu\nu} + \tilde{C}_{\mu\nu}$$

without FP ghost from diffeomorphism invariance.

◇ The effective action ( $m^2 = 0$ ):

$$S_{\text{Liouville}} = -\frac{N-25}{24\pi} \int_x \sqrt{-\hat{g}} (\hat{g}^{\mu\nu} \partial_\mu \sigma \partial_\nu \sigma + \hat{R} \sigma)$$

with  $N-25 \rightarrow N+1$ .

# Liouville action

- ◇ Liouville action with  $N \leq 25$  has kinetic terms with **right sign** due to the **FP ghost** contribution.

$$S_{\text{Liouville}} = -\frac{N-25}{24\pi} \int_x \sqrt{-\hat{g}} (\hat{g}^{\mu\nu} \partial_\mu \sigma \partial_\nu \sigma + \hat{R} \sigma)$$

- ◇ The FP ghost comes from gauge fixing of **diffeomorphism** invariance.
- ◇ What we need is **diffeomorphism invariance!**

This is consistent with the fact  $a = 0$  by regularizations which preserves diffeomorphism invariance.

(Recall:  $a = 0$  with Pauli-Villars reg. or Dimensional reg.)

# Generalization

- ◆ Above arguments are **valid** for **general** deformations of  $O(N)$  vector model with **arbitrary** potential  $V(\vec{\phi}^2)$  :

$$\frac{\mathcal{L}}{N} = \frac{1}{2}(\partial_\mu \vec{\phi})^2 - V(\vec{\phi}^2) + f(T, \tilde{T}_{\mu\nu})$$

( $f \dots$  **any** scalar function of  $T$  &  $\tilde{T}_{\mu\nu}$  )

- ◆ Because  $T = 2V(\vec{\phi}^2)$  , we can define

$$g(\vec{\phi}^2, \tilde{T}_{\mu\nu}) := V(\vec{\phi}^2) - f(T, \tilde{T}_{\mu\nu})$$

$$\frac{\mathcal{L}}{N} = \frac{1}{2}(\partial_\mu \vec{\phi})^2 - g(\vec{\phi}^2, \tilde{T}_{\mu\nu})$$

# Generalization

- ◆ This Lagrangian can be rewritten as

$$\frac{\mathcal{L}}{N} = \frac{1}{2} (\partial_\mu \vec{\phi})^2 - \frac{1}{2} C \vec{\phi}^2 - \frac{1}{2} \tilde{C}^{\mu\nu} \partial_\mu \vec{\phi} \cdot \partial_\nu \vec{\phi} + h(C, \tilde{C}^{\mu\nu})$$

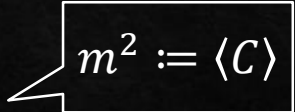
( $h \dots$  Legendre transformation of  $g$ )

- ◆ Effective action:

$$\frac{S_{\text{eff}}}{N} = \frac{i}{2} \text{tr} \log(-\partial^2 - C + \partial_\mu \tilde{C}^{\mu\nu} \partial_\nu) + \int_x h$$

- ◆ Gap equation:

$$\frac{1}{8\pi^2 i} \int_k \left( \frac{d^2 k}{k^2 - m^2} \right) + \frac{\partial h}{\partial C}(C = m^2, 0) = 0$$


$$m^2 := \langle C \rangle$$

# Generalization

◇ (1-loop) effective action:

$$\frac{S_{\text{eff}}}{N} = \frac{i}{2} \text{tr} \log(-\partial^2 - m^2 - \tilde{C} + \partial_\mu \tilde{C}^{\mu\nu} \partial_\nu) + \int_x h(m^2 + \tilde{C}, \tilde{C}^{\mu\nu})$$

◇  $h$  does not include derivatives of  $\tilde{C}$  &  $\tilde{C}^{\mu\nu}$

⇒ **kinetic terms** are **same** as the free case!

## Conclusion:

If deformation of a 2D theory by any function of  $T_{\mu\nu}$   
is well-defined at quantum level,  
the theory should have diffeomorphism invariance  
(or couple to gravity).

# Summary

- ◇ We studied the deformation of  $O(N)$  vector model in the large  $N$  limit, with  $1/N$  expansion.

$$\frac{\mathcal{L}}{N} = \frac{1}{2} (\partial_\mu \vec{\phi})^2 - \frac{m_0^2}{2} \vec{\phi}^2 + \alpha_0 \det(T_{\mu\nu})$$

1. Divergence can be **renormalized** (up to  $O(1/N)$  ).
2. **Positivity** of state space is **violated**.
3. **Diffeomorphism** invariance is **needed** to recover the positivity.
4. Above arguments are **valid** for **general** deformations with an **arbitrary** potential.

**Thank you!**

# Summary and Discussion

## ◆ Future direction

1. Considering **higher order**  $\left(\frac{1}{N}\right)^n$  **corrections**.

Can these save the violation of the positivity?

2. The **full solution** to the  $T\bar{T}$  flow equation for free  $O(N)$  vector Lagrangian is known. [Bonelli-Doroud-Zhu, Cavaglià-Negro-Szécsényi-Tateo]

$$\mathcal{L}^{(\alpha)} = \frac{1}{2\alpha} \left[ -1 + \sqrt{\left(1 + 2\alpha(\partial\vec{\phi} \cdot \bar{\partial}\vec{\phi})\right)^2 - 4\alpha^2(\partial\vec{\phi})^2(\bar{\partial}\vec{\phi})^2} \right]$$

(This takes the form of the Nambu-Goto action in  $D=N+2$  target space dimensions in the static gauge.)

Does higher order terms of  $\alpha$  change the results?

# Plan of talk

## ◇ Introduction

## ◇ $T\bar{T}$ deformation in large N limit

- $N \rightarrow \infty$  limit
- $\frac{1}{N}$  correction

## ◇ Summary and Discussion

# Back up

## ◇ Calculation of the effective action

$$\begin{aligned} \frac{S_{\text{eff}}}{N} = & -\text{tr} \left[ (-\partial^2 + (1 - 2\langle C \rangle)m^2)^{-1} (m^2 \tilde{C} + \partial_\mu \tilde{C}^{\mu\nu} \partial_\nu) \right]^2 \\ & + \int_x \frac{-1}{\alpha} \tilde{C}^2 + \frac{1}{2\alpha} (\tilde{C}^{\mu\nu})^2 \end{aligned}$$

# 素材

